Solution 1

Solution 1 (a)

The function $\phi(x)$ returns the index j corresponding to the closest cluster center. Since there are k cluster centers indexed from 1 to k, the output is a single discrete integer.

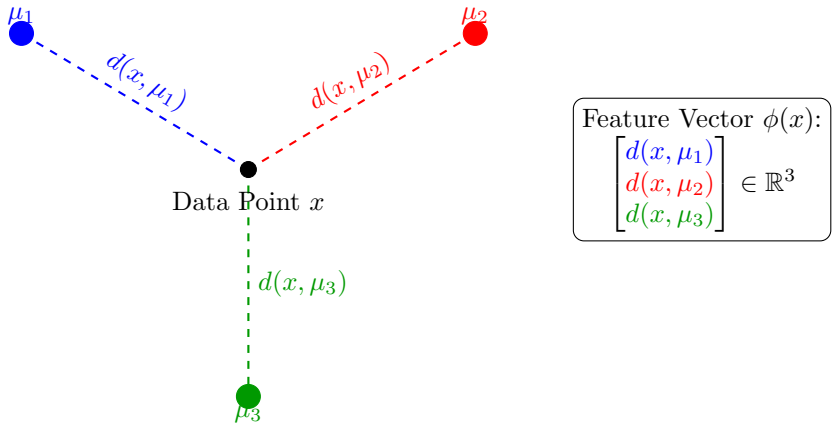
$\therefore$ Thus, the space is the discrete set of indices:

$$\phi(x) \in \{1, 2, \ldots, k\}$$

Solution 1

Solution 1 (b)

Since distances in a metric space are non-negative real numbers, $\phi(x)$ is a vector in a k-dimensional Euclidean space (specifically the non-negative orthant).



$\therefore \phi(x) \in \mathbb{R}^k$

Solution 2

---

1. Case 1: Matrix D_1

   From the matrix, the distances are:

   $$d(x_1, x_2) = 1, \quad d(x_2, x_3) = 1, \quad d(x_1, x_3) = 2$$

   Now, check the triangle inequality:

   $$d(x_1, x_2) \leq d(x_1, x_2) + d(x_2, x_3)$$

   $$d(x_1, x_2) = 1 \quad \text{and} \quad d(x_1, x_2) + d(x_2, x_3) = 1 + 1 = 2$$

   From above, the triangle inequality holds.
   We can construct these points in 1-dimensional Euclidean space ($\mathbb{R}^1$) by placing them on the number line:

   $$x_1 = 0, \quad x_2 = 1, \quad x_3 = 2$$

2. Case 2: Matrix D_2

   From the matrix, the distances are:

   $$d(x_1, x_2) = 1, \quad d(x_2, x_3) = 1, \quad d(x_1, x_3) = 3$$

   Now, check the triangle inequality:

   $$d(x_1, x_3) \leq d(x_1, x_2) + d(x_2, x_3)$$

   $$d(x_1, x_3) = 3 \quad \text{and} \quad d(x_1, x_2) + d(x_2, x_3) = 1 + 1 = 2$$

   $$d(x_1, x_3) \not\leq d(x_1, x_2) + d(x_2, x_3)$$

$\therefore$ D_1 is realizable ($x_1 = 0, x_2 = 1, x_3 = 2$) and D_2 is not realizable.

Solution 3

Solution 3 (a)

Let the points be denoted as vectors:

$$x = \begin{bmatrix}1\\2\\3\end{bmatrix}, \quad y = \begin{bmatrix}-1\\0\\1\end{bmatrix}, \quad z = \begin{bmatrix}0\\-2\\-4\end{bmatrix}$$

The matrix D consists of squared Euclidean distances $D_{ij} = ||x_i - x_j||^2$ or can be written as:

$$\begin{bmatrix}||x_1 - x_1||^2 & ||x_1 - y_1||^2 & ||x_1 - z_1||^2\\ ||x_2 - x_1||^2 & ||y_2 - y_2||^2 & ||x_2 - z_2||^2\\ ||x_3 - x_1||^2 & ||x_1 - y_1||^2 & ||x_3 - z_3||^2\end{bmatrix}$$

Now calculate each distance element:

1. D_{11}
$$||x - x||^2 = (1-1)^2 + (2-2)^2 + (3-3)^2 = 0^2 + 0^2 + 0^2 = 0$$
2. D_{12}
$$||x - y||^2 = (1-(-1))^2 + (2-0)^2 + (3-1)^2 = 2^2 + 2^2 + 2^2 = 4 + 4 + 4 = 12$$
3. D_{13}
$$||x - z||^2 = (1-0)^2 + (2-(-2))^2 + (3-(-4))^2 = 1^2 + 4^2 + 7^2 = 1 + 16 + 49 = 66$$
4. D_{21}
$$||y - x||^2 = (-1-1)^2 + (0-2)^2 + (1-3)^2 = (-2)^2 + (-2)^2 + (-2)^2 = 4 + 4 + 4 = 12$$
5. D_{22}
$$||y - y||^2 = (-1-(-1))^2 + (0-0)^2 + (1-1)^2 = 0^2 + 0^2 + 0^2 = 0$$
6. D_{23}
$$||y - z||^2 = (-1-0)^2 + (0-(-2))^2 + (1-(-4))^2 = (-1)^2 + 2^2 + 5^2 = 1 + 4 + 25 = 30$$
7. D_{31}
$$||z - x||^2 = (0-1)^2 + (-2-2)^2 + (-4-3)^2 = (-1)^2 + (-4)^2 + (-7)^2 = 1 + 16 + 49 = 66$$
8. D_{32}
$$||z - y||^2 = (0-(-1))^2 + (-2-0)^2 + (-4-1)^2 = 1^2 + (-2)^2 + (-5)^2 = 1 + 4 + 25 = 30$$
9. D_{33}
$$||z - z||^2 = (0-0)^2 + (-2-(-2))^2 + (-4-(-4))^2 = 0$$

$$\therefore D = \begin{bmatrix}0 & 12 & 66\\ 12 & 0 & 30\\ 66 & 30 & 0\end{bmatrix}$$

Solution 3

Solution 3 (b)

H is the centering matrix defined as:

$$H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$$

,

Let, $n = 3$ and I is the identity matrix. Now, calculate H

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{3}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$$

$$\therefore H = \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix}$$

Solution 3

Solution 3 (c)

This operation converts the squared distance matrix into the centered Gram matrix B.

$$B = -\frac{1}{2}HDH$$

Performing the matrix multiplication: 1. $DH = \begin{bmatrix} 0 & 12 & 66 \\ 12 & 0 & 30 \\ 66 & 30 & 0 \end{bmatrix} \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix} = \begin{bmatrix} -26 & -14 & 40 \\ -2 & 14 & -12 \\ 34 & -12 & -22 \end{bmatrix}$

2. $H(DH) = \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix} \begin{bmatrix} -26 & -14 & 40 \\ -2 & 14 & -12 \\ 34 & -12 & -22 \end{bmatrix} = \begin{bmatrix} -28 & -4 & 32 \\ -4 & -4 & 8 \\ 32 & 8 & -40 \end{bmatrix}$

3. Multiply by $-\frac{1}{2}$:

$$B = -\frac{1}{2} \begin{bmatrix} -28 & -4 & 32 \\ -4 & -4 & 8 \\ 32 & 8 & -40 \end{bmatrix}$$

$\therefore \begin{bmatrix} 14 & 2 & -16 \\ 2 & 2 & -4 \\ -16 & -4 & 20 \end{bmatrix}$

Solution 3

Solution 3 (d)

The definition of the Gram matrix for a set of data vectors (arranged as rows in X) is $G = XX^T$. This means the entry G_{ij} is the dot product of the i-th point and the j-th point.

Let,the data matrix X be defined as:

$$X = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 0 & -2 & -4 \end{bmatrix}$$

The rows correspond to vectors x, y, z. We calculate the dot products directly:

$$\begin{aligned} x \cdot x &= (1)(1) + (2)(2) + (3)(3) = 1 + 4 + 9 = 14 \\ y \cdot y &= (-1)(-1) + (0)(0) + (1)(1) = 1 + 0 + 1 = 2 \\ z \cdot z &= (0)(0) + (-2)(-2) + (-4)(-4) = 0 + 4 + 16 = 20 \\ x \cdot y &= (1)(-1) + (2)(0) + (3)(1) = -1 + 0 + 3 = 2 \\ x \cdot z &= (1)(0) + (2)(-2) + (3)(-4) = 0 - 4 - 12 = -16 \\ y \cdot z &= (-1)(0) + (0)(-2) + (1)(-4) = 0 + 0 - 4 = -4 \end{aligned}$$

By symmetry, $x \cdot y = y \cdot x$, etc. Constructing the matrix from these values:

$$XX^T = \begin{bmatrix} 14 & 2 & -16 \\ 2 & 2 & -4 \\ -16 & -4 & 20 \end{bmatrix}$$

This result matches the matrix B calculated in part (c) exactly. *Note: The operation in (c) produces the Gram matrix of the centered data. Since the mean of the original points is $\vec{0}$, the centered Gram matrix and the original Gram matrix are identical.*

$\therefore$ Yes, the result matches the matrix from part (c).

Solution 4

Solution 4 (a)

Let the points be vectors $x_1, \ldots, x_n$. The definition of **orthonormal** is: 1. Orthogonal: The dot product of distinct vectors is zero ($x_i^T x_j = 0$ for $i \neq j$). 2. Normal: The length of each vector is one ($x_i^T x_i = ||x_i||^2 = 1$).

The Gram matrix G is defined as $G_{ij} = x_i^T x_j$. Applying the definition:

- Diagonal entries ($i = j$): $G_{ii} = x_i^T x_i = 1$.
- Off-diagonal entries ($i \neq j$): $G_{ij} = x_i^T x_j = 0$.

Thus, G is the Identity matrix.

$\therefore G = I_n$ (The $n \times n$ Identity Matrix)

Solution 4

Solution 4 (b)

The squared Euclidean distance between two vectors x_i and x_j is given by the expansion:

$$||x_i - x_j||^2 = (x_i - x_j)^T(x_i - x_j)$$
$$= x_i^T x_i - 2x_i^T x_j + x_j^T x_j$$

Substitute the orthonormal properties from part (a):

- If $i = j$: $||x_i - x_i||^2 = 0$.
- If $i \neq j$: $||x_i - x_j||^2 = 1 - 2(0) + 1 = 2$.

Therefore, the squared distance matrix D has 0 on the diagonal and 2 everywhere else.

$$\therefore D = 2(\mathbf{1}\mathbf{1}^T - I) = \begin{bmatrix} 0 & 2 & \dots & 2 \\ 2 & 0 & \dots & 2 \\ \vdots & \vdots & \ddots & \vdots \\ 2 & 2 & \dots & 0 \end{bmatrix}$$

Solution 5

Solution 5 (a)

Python Code

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import euclidean_distances

def cmdscale(D_squared, k=2):
    """ implement cMDS from lecture"""
    n = D_squared.shape[0]

    # calc centering matrix H
    H = np.eye(n) - (1/n) * np.ones((n, n))

    # calce B = -1/2 * H * D_squared * H
    B = -0.5 * H @ D_squared @ H

    # spectral decomposition
    # note: np.linalg.eigh returns eigenvalues in ascending order
    evals, evecs = np.linalg.eigh(B)

    # sort
    idx = np.argsort(evals)[::-1]
    evals = evals[idx]
    evecs = evecs[:, idx]

    # 4. zero negative eigenvalues (for non-Euclidean noise)
    evals_pos = np.maximum(evals, 0)

    # calc coordinates Y
    Lambda_sqrt = np.diag(np.sqrt(evals_pos))

    # project: Y = U * Lambda^1/2
    Y = evecs @ Lambda_sqrt

    # return first k columns
    return Y[:, :k]

# read data
raw_data = open('cities.txt').read().split()
city_names = raw_data[2:12]

# massage data
matrix_values = [float(x) for x in raw_data[12:]]
D = np.array(matrix_values).reshape(10, 10)
D_squared = D ** 2

# run cmdscale(D_squared, k=2) function
Y_k = cmdscale(D_squared, k=2)

def rotate(points, angle_degrees):
    theta = np.radians(angle_degrees)
    c, s = np.cos(theta), np.sin(theta)
    R = np.array([[c, -s], [s, c]])
    return points @ R.T

# rotate 180
Y_rotated = rotate(Y_k, angle_degrees=180)

# plot
plt.figure(figsize=(10, 6))
plt.scatter(Y_rotated[:, 0], Y_rotated[:, 1], c='blue', s=100)
for i, name in enumerate(city_names):
    plt.annotate(name, (Y_rotated[i, 0], Y_rotated[i, 1]),
                 xytext=(5, 5), textcoords='offset points')

```

```
plt.title("Reconstructed Map from Distance Matrix (MDS)")
plt.grid(True)
plt.axis('equal')
plt.savefig("q5a.png")
```

Plot

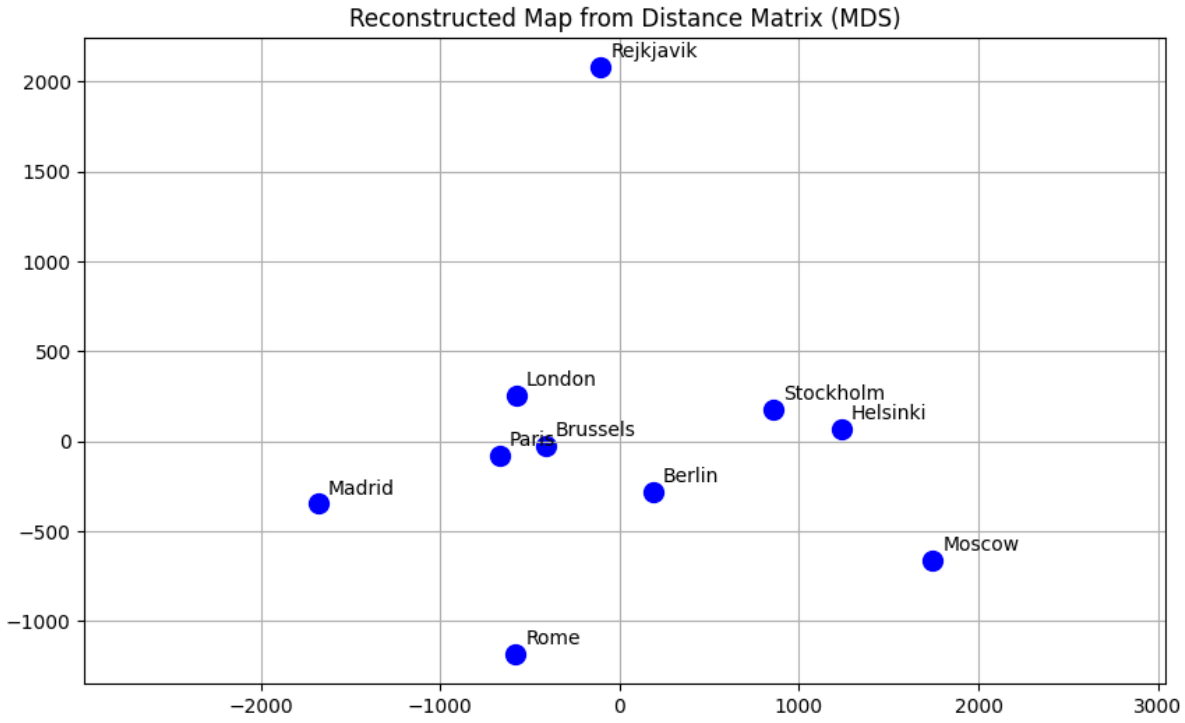


Figure 1: Plot of cities.txt

Solution 5

Solution 5 (b)

Python Code

```python
def cmdscale(D, k=2):
    """classicMDS: Projects n points into k-dimensional space."""
    n = D.shape[0]
    H = np.eye(n) - (1/n) * np.ones((n, n))
    B = -0.5 * H @ D @ H

    evals, evecs = np.linalg.eigh(B)

    # sort
    idx = np.argsort(evals)[::-1]
    evals = evals[idx]
    evecs = evecs[:, idx]

    # project
    w_pos = np.maximum(evals, 0)
    Y = evecs @ np.diag(np.sqrt(w_pos))
    return Y[:, :k]

def rotate_data(X, angle_degrees):
    """Rotate 2D data X by angle_degrees (counter-clockwise)"""
    theta = np.radians(angle_degrees)
    c, s = np.cos(theta), np.sin(theta)

    # define rotation matrix
    R = np.array([[c, -s], [s,  c]])

    # apply R to every point
    return X @ R.T

# plot
def plot_map(X, labels, title="MDS Map"):
    plt.figure(figsize=(8, 8))
    plt.scatter(X[:, 0], X[:, 1], c='red', marker='o')
    for i, txt in enumerate(labels):
        plt.annotate(txt, (X[i, 0], X[i, 1]), xytext=(5, 5), textcoords='offset points')

    plt.title(title)
    plt.grid(True)
    plt.axis('equal')
    plt.show()

# get raw MDS coordinates
Y_raw = cmdscale(D_squared)

# rotate
best_angle = 190
Y_rotated = rotate_data(Y_raw, best_angle)

# call plot_map function
plot_map(Y_rotated, city_names, title=f"European Cities (Rotated {best_angle} degrees)")
```

Plot

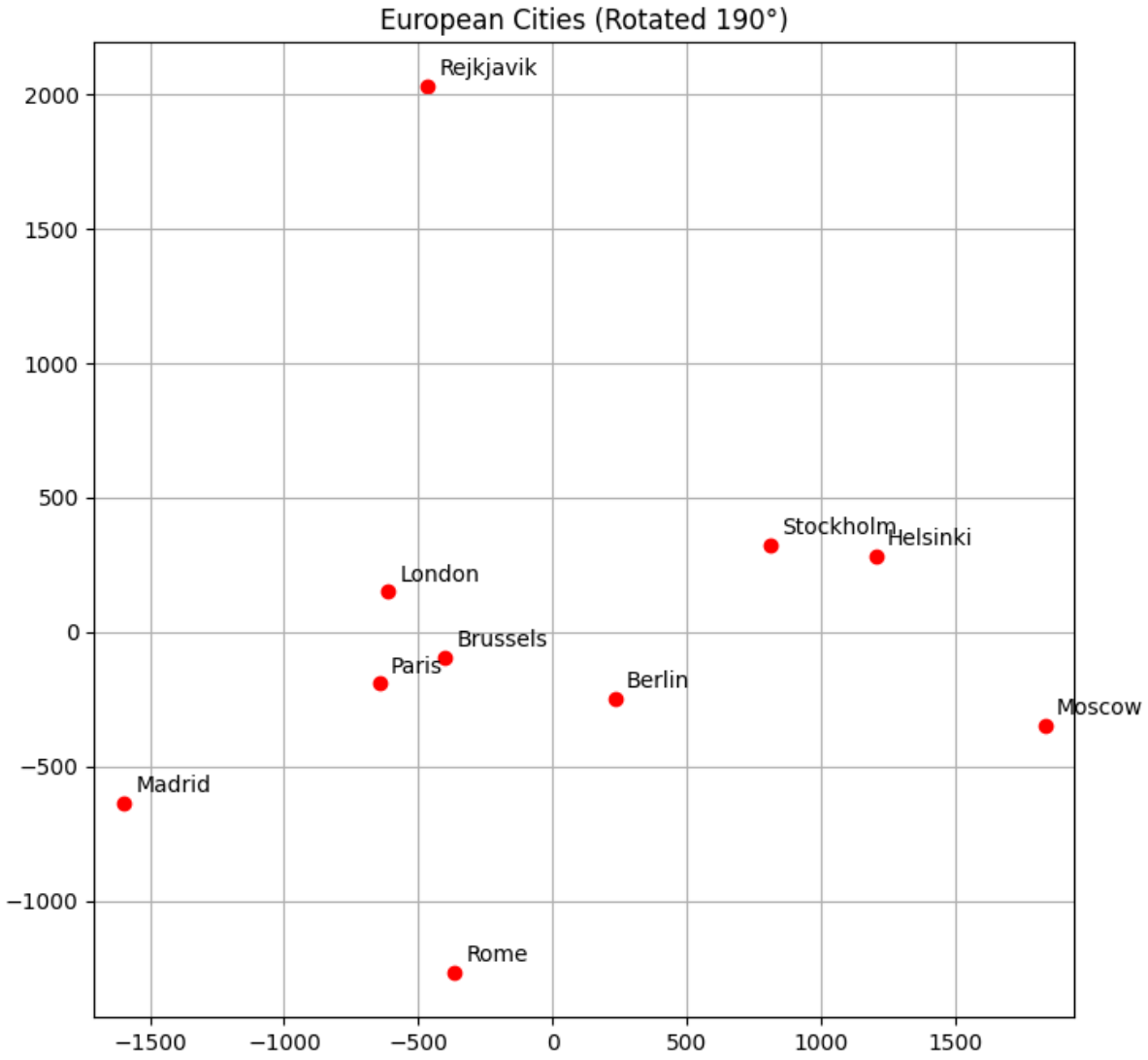


Figure 2: Plot 5b

Solution 5

Solution 5 (c)

Python Code

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.manifold import MDS
# sklearn MDS
mds = MDS(n_components=2, dissimilarity='precomputed', random_state=42,
    normalized_stress='auto')
Y_sklearn = mds.fit_transform(D) #
# rotation
def rotate_data(X, angle_degrees):
    theta = np.radians(angle_degrees)
    c, s = np.cos(theta), np.sin(theta)
    R = np.array([[c, -s], [s,  c]])
    return X @ R.T

def plot_embedding(X, title, ax):
    ax.scatter(X[:, 0], X[:, 1], c='green', s=100)
    for i, txt in enumerate(city_names):
        ax.annotate(txt, (X[i, 0], X[i, 1]), xytext=(5, 5), textcoords='offset points')
    ax.set_title(title)
    ax.grid(True)
    ax.axis('equal')

# find correct rotation
angle = 150
Y_rotated = rotate_data(Y_sklearn, angle)
# plot
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 8))
plot_embedding(Y_sklearn, "Sklearn MDS (Raw Output)", ax1)
plot_embedding(Y_rotated, f"Sklearn MDS (Rotated {angle} degrees)", ax2)
plt.savefig("q5c.png")
```

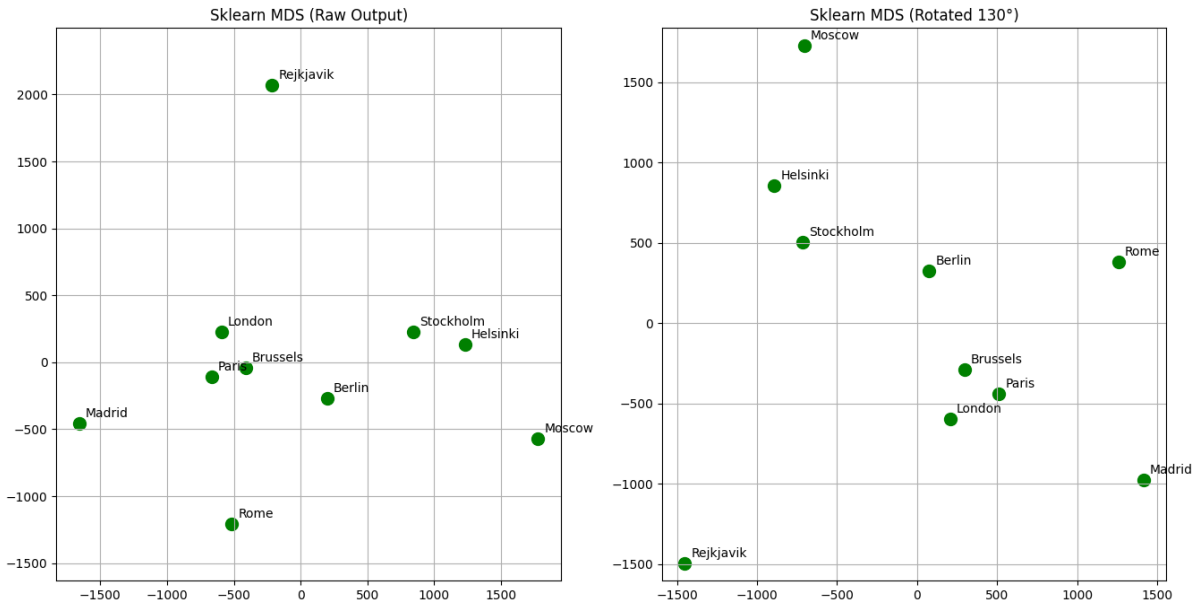


Figure 3: Plot 5 c